

# Gabriel Mankowski

gabemankowski11@gmail.com | Garretson, SD | (952) 956 2048

## EDUCATION

|                                                                                                         |                          |
|---------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Iowa State University</b>                                                                            | <b>Expected May 2027</b> |
| <ul style="list-style-type: none"><li>Bachelor of Science in Computer Engineering (GPA: 3.85)</li></ul> |                          |

## SKILLS

- Programming Languages: Python, C, C++, Java, SQL
- Hardware & Tools: Verilog, VHDL, Microchip IGLOO nano, Libero SoC, QuestaSim, Quartus Prime, MSP430, TM4C, RISC-V, Git, Linux
- Cloud & Frameworks: AWS (boto3, S3, Step Functions), FastAPI, Spring Boot, WebSockets

## EXPERIENCE

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| <b>Software Engineer Intern   Innovate!, Inc. (Subcontractor to KBR / USGS TSSC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>May 2026 - Present</b>  |
| <ul style="list-style-type: none"><li>Engineered asynchronous Python services using FastAPI and AWS SDK (boto3) to track cloud state machine execution and S3 pipeline data transfers for the Annual NLCD.</li><li>Developed automated integration and unit test suites to validate data processing workflows across distributed cloud storage environments.</li><li>Collaborated with core engineering teams to optimize cloud data retrieval pipelines, decreasing manual monitoring overhead.</li></ul> |                            |
| <b>Undergraduate Research Assistant   Iowa State University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Jan 2026 - May 2026</b> |
| <ul style="list-style-type: none"><li>Designed an ultra-low-power ML inference accelerator in VHDL for a Microchip IGLOO nano FPGA using Libero SoC.</li><li>Synthesized parallel MAC units, variable quantization (2/4/8-bit), and memory banking to execute LeNet-5 and FastInf models under strict logic cell constraints.</li><li>Verified pipeline timing and resolved gate-level propagation delays using QuestaSim/ModelSim simulation testbenches.</li></ul>                                       |                            |
| <b>Teaching Assistant   Dept. of Electrical &amp; Computer Engineering</b>                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Aug 2025 - Present</b>  |
| <ul style="list-style-type: none"><li>Computer Organization &amp; Assembly (CPRE 3810): Guide students in pipelined RISC-V processor design, hazard mitigation, and cache memory.</li><li>Digital Logic (CPRE 2810): Mentored 45+ students weekly on finite state machines, combinational/sequential logic, and Verilog simulation.</li></ul>                                                                                                                                                              |                            |

## PROJECTS

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| <b>Obstacle-Avoiding Roomba with IR and Ultrasonic Sensors</b>                                                                                                                                                              | <b>Jan 2025 - May 2025</b> |
| <ul style="list-style-type: none"><li>Programmed a TM4C microcontroller to scan environments with IR and PING sensors.</li><li>Developed algorithms for detecting narrow objects and navigating around obstacles.</li></ul> |                            |

## ACTIVITIES AND LEADERSHIP

|                                        |                     |
|----------------------------------------|---------------------|
| Controlled Environments Club President | Sep 2023 - May 2026 |
| ISU Datathon 3rd place                 | October 2023        |
| Eagle Scout                            |                     |